Google Cloud (GCP) Implementation - Big Data, GenAI & Agentic AI Architecture

Data sources

Data ingestion

Batch data
Storage Transfer Service
BigQuery Data Transfer · Composer

Real-time streaming
Cloud Pub/Sub
Dataflow (Apache Beam)

Change data capture
Datastream · DMS
Cloud SQL → BigQuery

Reverse ETL
Dataflow · Workflows
Eventarc · Cloud Functions

Data Ocean

Data lake
Cloud Storage + BigLake

Data pond
GCS domain bucket + Dataplex

Data puddle
GCS project bucket + IAM

Data mesh
Dataplex · Analytics Hub

Data contracts
Dataplex Catalog · Pub/Sub Schema

Workflow manager · Cloud Composer (Airflow) · Workflows · Cloud Scheduler · Eventarc

Data transformation

Data labeling
Vertex AI Data Labeling

Query engines
BigQuery · Dataproc (Spark) · BigLake

Data sci. libraries
Vertex AI Workbench · Colab Enterprise

Data enrichment
Analytics Hub · Dataflow

AI infrastructure

Knowledge store

Feature store
Vertex AI Feature Store

Vector database
Vertex AI Vector Search
AlloyDB / Cloud SQL pgvector

Knowledge graph
Spanner Graph · Neo4j on GCE

LLMOps + MLOps

Model registry
Vertex AI Model Registry

Prompt registry
Vertex AI Prompt Management

Fine-tuning · RLHF
Vertex AI Tuning · Model Garden

Eval & red-teaming
Vertex AI Evaluation · Model Armor

CI/CD · experiment tracking
Cloud Build · Deploy · Vertex Pipelines

AI inference

Online predictor
Vertex AI Online Prediction Endpoints

Batch predictor
Vertex AI Batch Prediction · Dataproc

Streaming inference
Dataflow + Vertex AI · Cloud Functions

Edge inference
Edge TPU · Coral · Distributed Cloud Edge

AI gateway
Apigee · Cloud Endpoints · API Gateway

GenAI platform

Guardrails & orchestration
Model Armor · Vertex AI Safety Filters

Model & prompt hub
Vertex AI Model Garden · GenAI Studio

Custom knowledge model
Vertex AI Custom Training · Model Garden

Foundation models
Vertex AI Model Garden
Gemini · Imagen · Codey · Llama · Claude

RAG pipeline
Vertex AI RAG Engine
Vector Search · AlloyDB pgvector

Multimodal
Gemini Multimodal · Vision API
Speech-to-Text · Document AI

Agentic AI layer

Agent orchestrator
Vertex AI Agent Builder
Agent Development Kit (ADK)
Workflows

Tool registry / MCP
Vertex AI Extensions
Cloud Functions · Cloud Run

Agent memory
Vertex AI Memory Bank
Firestore · Memorystore
Vector Search

HITL interface
Vertex AI Human-in-the-Loop
Data Labeling Service

Agent monitoring
Cloud Trace · Cloud Logging
Vertex AI Model Monitoring

Analytics & BI

Data viz & storyboard
Looker · Looker Studio

Data search
Vertex AI Search · Discovery Engine

Data accumulation
BigQuery · Spanner · Memorystore

Content analysis
Natural Language API · Document AI
Translation · Speech-to-Text

Decision management
Workflows · Vertex AI
Recommendations AI

Agentic analytics
Gemini in Looker
Gemini in BigQuery · Duet AI

AI trust, governance & security

Data governance
Dataplex · Data Catalog · BigQuery Governance
Cloud DLP (Sensitive Data Protection) · Cloud KMS
Cloud Audit Logs · IAM · VPC Service Controls · SCC

AI governance
Vertex AI Model Cards · Model Monitoring · Model Armor
Responsible AI Toolkit · Cloud Audit Logs
Vertex AI Explainable AI · Safety Filters

Assessment framework

AI maturity assessment
Cloud Architecture Framework · Active Assist

Economic benefit assessment
Cloud Billing · Pricing Calculator · Budgets

Responsible AI KPIs
Vertex AI Model Monitoring · Explainable AI
Cloud Monitoring · Vertex AI Evaluation

Applications

Data layers | AI/ML infrastructure | GenAI & analytics | Agentic AI | Governance & trust | Assessment